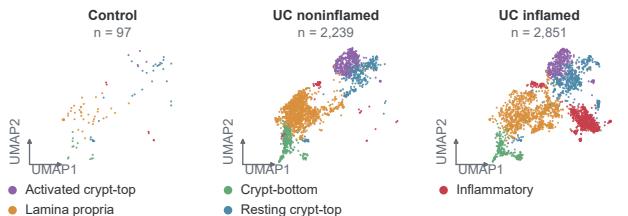
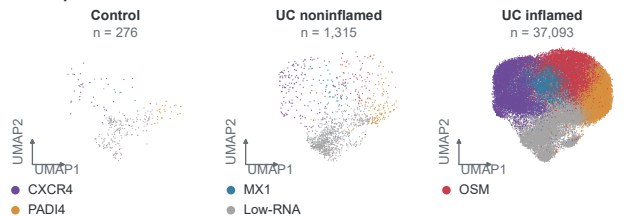


a Human UC fibroblast and neutrophil state landscape

Fibroblasts



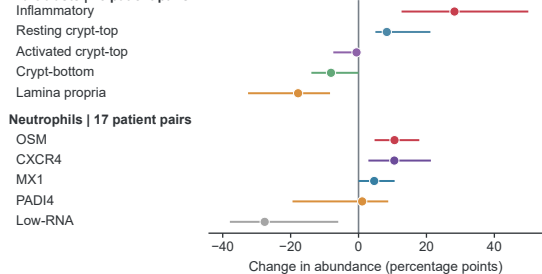
Neutrophils



b Paired cell-state remodeling

Inflamed minus noninflamed; median and 95% CI

Fibroblasts | 13 patient pairs

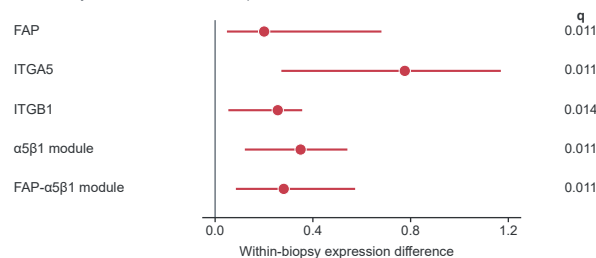


Neutrophils | 17 patient pairs



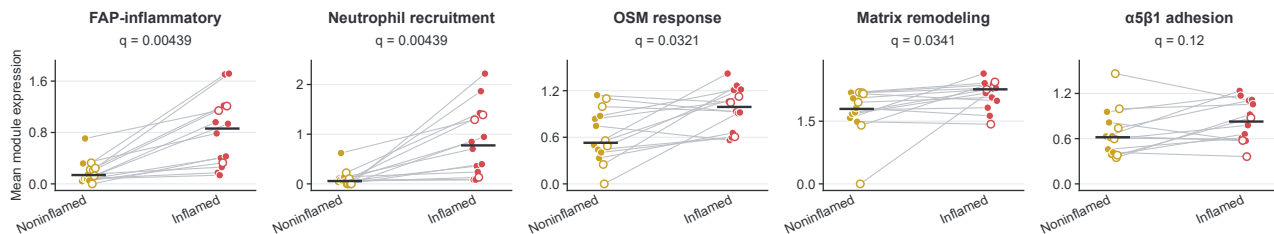
c Inflammatory-fibroblast activation

Inflammatory vs other fibroblasts; 10 biopsies



d Paired fibroblast program changes

14 patient pairs; lines connect patients; black bars, medians; open points, <10 recovered cells



e Neutrophil state-associated RNA programs

11 matched patients; ≥20 cells/state; naming genes excluded



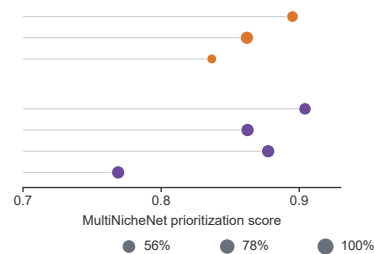
f Predicted reciprocal signals

Fibroblast → neutrophil

HGF → CD44
CXCL1 → CXCR2
CSF3 → CSF3R

Neutrophil → fibroblast

OSM → OSMR
IL1B → IL1R1
NAMPT → ITGA5
NAMPT → ITGB1



Equal patient weights; Friedman q (BH, 9 programs).

Paired coexpression: Extended Data Fig. 1c.